Practice Session 6 Answers

Part 1: Hypothesis Testing Terms and Definitions

Generally speaking, hypothesis testing is a way for researchers to answer questions in a meaningful way. The methodology leverages statistical reasoning and probability to either reject or fail to reject a statement about the population as a whole. Hypothesis testing in statistics often focuses on questions about population parameters (e.g., μ, π). We do not write hypotheses in terms of the sample statistics (e.g., $\bar{x}$, $\hat{p}$). This aligns with the general theme of the course, which focuses on using samples to better understand the population.

Practice 1:

Please match the following terms with the appropriate definition.

Terms:

1. Null Hypothesis
2. Alternative Hypothesis
3. P-value

Definitions:

a. The probability of observing results at least as extreme as what we observed, assuming the null hypothesis is true.
b. A statement of interest that represents the status quo, or that there is no effect or difference.
c. A statement of interest that proposes there is an effect or difference.

Answers

1. **Null Hypothesis:** A statement of interest that represents the status quo, or that there is no effect or difference.
2. **Alternative Hypothesis:** A statement of interest that proposes there is an effect or difference.

3. **P-value:** The probability of observing results at least as extreme as what we observed, assuming the null hypothesis is true.

Part 2: Hypothesis Testing for a Single Proportion

Practice 2.1: Preferred Water Brand

A national sales bureau believes that 30% of consumers prefer Fiji water as their first choice for water. Specifically, they choose it over Aquafina, Sam's Choice, and tap water. The Fiji company claims that this proportion is greater than 30%. Using the `WaterTaste` data set from the `Lock5Data` library, run a hypothesis test to check this claim. Look at the help page for the data set to see which variable is appropriate to use.

```
library(SDS1000)
library(Lock5Data)
data(WaterTaste)
```

Answers

1. **First**: State the null and alternative hypotheses in words and in symbols.
    - Null: The proportion of consumers who choose Fiji as their first choice is 0.30.
    - Alternative: The proportion of consumers who choose Fiji as their first choice is greater than 0.30.
    - $H_0 : \pi = 0.3$
    - $H_a : \pi > 0.3$
2. **Second**: Calculate the observed statistic of interest.

```
p_hat_fiji <- get_proportion(WaterTaste$First, "Fiji")
p_hat_fiji
```

```
Fiji
0.41
```

3. **Third**: Create a null distribution in R using `do_it()` and `rflip()`. Plot the null distribution, and add a vertical red line at your point estimate.

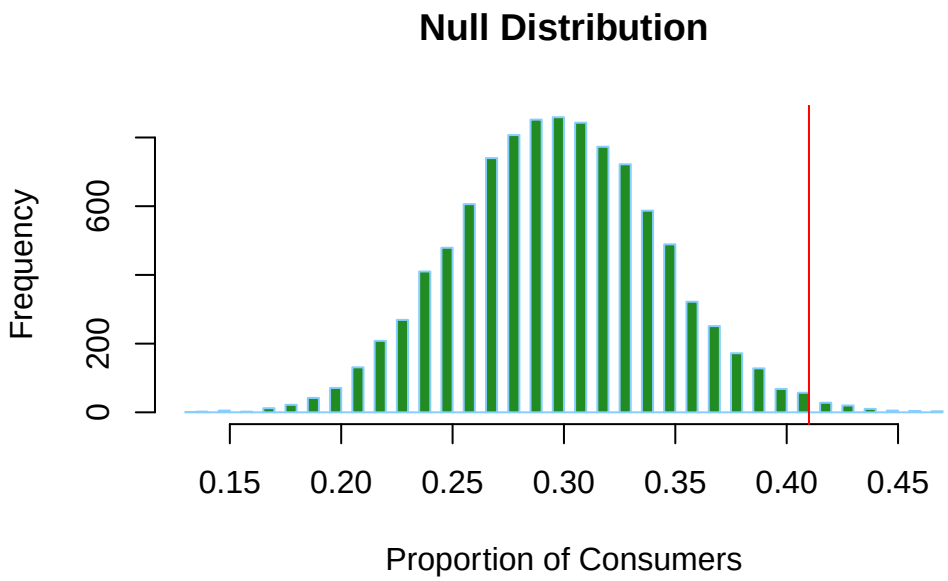
```r
# Number of consumers to simulate
n_water <- 100

# Set the seed so everyone gets the same simulated results
set.seed(1000)

# Simulate the null distribution
null_dist_fiji <- do_it(10000) * {
  rflip(n_water, 0.3) / n_water
}

# Another way to create the null distribution
# null_dist_fiji <- do_it(10000) * {
#   rflip(n_water, 0.3, report_proportion = TRUE)
# }

# Visualize the null distribution
hist(null_dist_fiji, breaks = 50,
     main = "Null Distribution", xlab = "Proportion of Consumers",
     col = "forestgreen", border = "skyblue1")
abline(v = p_hat_fiji, col = "red")
```



4. **Fourth**: Calculate the p-value (probability of obtaining a result as or more extreme

than what we observed) using the `ptail()` function.

```
one_sided_pv <- ptail(p_hat_fiji, null_dist_fiji, lower.tail = FALSE)
one_sided_pv
```

```
[1] 0.0127
```

5. **Fifth**: Report the p-value. Make a decision on whether the results are statistically significant, and state your conclusion.

Since our p-value is less than the standard 0.05, we will reject the null hypothesis. We therefore have evidence that over 30% of consumers would select Fiji water as their first choice of water.

Practice 2.2: AP Multiple Choice

Does the answer choice "C" occur less frequently than expected? Answer this question with a hypothesis test. Use the `APMultipleChoice` data set from the `Lock5Data` library. Make sure to follow all 5 steps of hypothesis testing. *Hint*: when defining the null hypothesis, consider how likely each answer choice would be if answers were selected at random.

```
library(SDS1000)
library(Lock5Data)
data(APMultipleChoice)
```

Answers

1. **First**: State the null and alternative hypotheses in words and in symbols.
    - Null: The proportion of questions with "C" as the correct answer is 0.2 (what we would expect if each of the five choices were equally likely).
    - Alternative: The proportion of questions with "C" as the correct answer is less than 0.2.
    - $H_0 : \pi = 0.2$
    - $H_a : \pi < 0.2$
2. **Second**: Calculate the observed statistic of interest.

```r
p_hat_ap <- get_proportion(APMultipleChoice$Answer, "C")
p_hat_ap
```

```
     C
0.1975
```

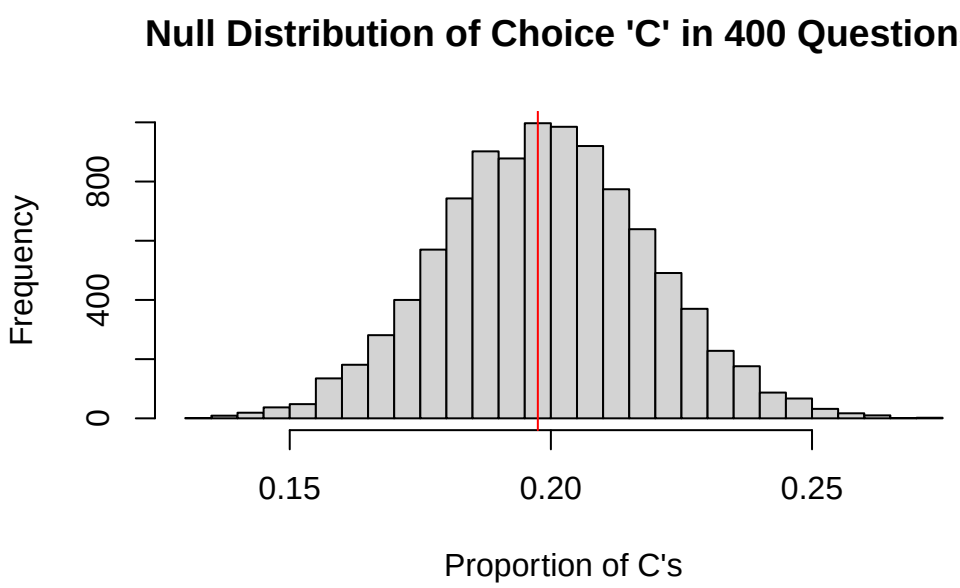
3. **Third**: Create a null distribution in R using `do_it()` and `rflip()`. Plot the null distribution, and add a vertical red line at your point estimate. *Note*: there were 400 multiple-choice questions in the data set.

```r
# Number of questions to simulate
n_questions <- 400

# Set the seed so everyone gets the same simulated results
set.seed(1000)

# Simulate the null distribution
null_dist_ap <- do_it(10000) * {
  rflip(n_questions, 0.2) / n_questions
}

# Visualize the null distribution
hist(null_dist_ap, breaks = 50,
     main = "Null Distribution of Choice 'C' in 400 Questions",
     xlab = "Proportion of C's")
abline(v = p_hat_ap, col = "red")
```



4. **Fourth**: Calculate the p-value (probability of obtaining a result as or more extreme than what we observed) using the `ptail()` function.

```
one_sided_pv <- ptail(p_hat_ap, null_dist_ap, lower.tail = TRUE)
one_sided_pv
```

```
[1] 0.4721
```

5. **Fifth**: Report the p-value. Make a decision on whether the results are statistically significant, and state your conclusion.

Since our p-value is greater than the standard 0.05, we will fail to reject the null hypothesis. We therefore do not have evidence that the answer choice "C" occurs less frequently than expected.

Part 3: More practice/ UFO Shapes

UFOs, or Unidentified Flying Objects, have become commonplace in media and popular culture. Some witnesses claim that the objects have a cylindrical shape, while others believe they are more spherical. Load the data `ufo_data.csv` using the function `read.csv()`. The file

is in the course materials folder; save it to your working directory (the same folder as this file) before running `read.csv()`. The data have already been filtered to include only sightings described as spheres or cylinders.

Run a hypothesis test to check whether the proportion of sightings described as spherical differs from 0.75.

Answers:

1. State the null and alternative hypotheses in words and in symbols.
   - Null: The proportion of UFO sightings described as spherical is 0.75.
   - Alternative: The proportion of UFO sightings described as spherical is different from 0.75.
   - $$H_0 : \pi = 0.75$$
   - $H_a : \pi \neq 0.75$
2. Calculate the observed statistic of interest.

```
library(SDS1000)
ufo_data <- read.csv("ufo_data.csv")

p_hat_sphere <- get_proportion(ufo_data$shape, "sphere")
p_hat_sphere
```

```
   sphere
0.8076462
```

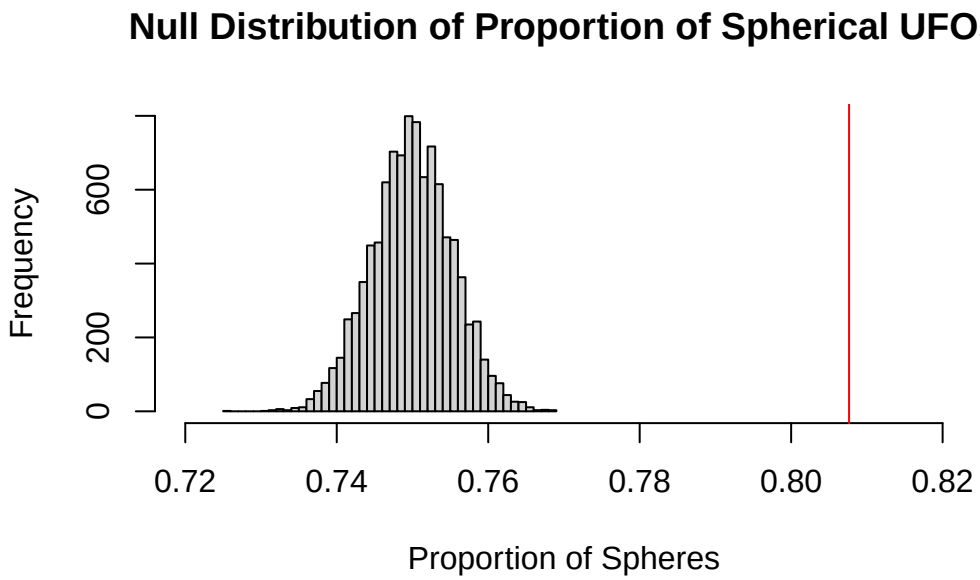
3. Create a null distribution in R using `do_it()` and `rflip()`. Plot the null distribution, and add a vertical red line at your point estimate.

```
# Number of UFO sightings to simulate
n_ufos <- 6670

# Set the seed so everyone gets the same simulated results
set.seed(1000)

# Simulate the null distribution
null_dist_sphere <- do_it(10000) * {
  rflip(n_ufos, 0.75) / n_ufos
}
```

```r
# Visualize the null distribution
hist(null_dist_sphere, breaks = 50,
     main = "Null Distribution of Proportion of Spherical UFOs",
     xlab = "Proportion of Spheres", xlim = c(0.72, 0.82))
abline(v = p_hat_sphere, col = "red")
```



4. Calculate the p-value (probability of obtaining a result as or more extreme than what we observed) using the `ptail()` function.

Because this is a two-sided test, "as or more extreme" means as far from the null value **in either direction**. Our observed proportion is about 0.058 above 0.75, so a result 0.058 *below* 0.75 would be just as extreme. We find that distance, mark a cutoff that far out on each side of the null value, and add the two tail probabilities.

```r
null_val <- 0.75

# Distance between the observed proportion and the null value
margin <- abs(p_hat_sphere - null_val)

# Upper tail: as far above 0.75 as we observed, or farther
upper <- ptail(null_val + margin, null_dist_sphere, lower.tail = FALSE)
```

```r
# Lower tail: the same distance below 0.75, or farther
lower <- ptail(null_val - margin, null_dist_sphere, lower.tail = TRUE)

pv <- lower + upper
pv
```

```
[1] 0
```

None of our 10,000 simulated proportions were as extreme as the one we observed, so `ptail()` returns 0. That does **not** mean the p-value is exactly 0 (that would mean our result is impossible under the null). A simulation-based p-value can't go below 1 / (number of simulations), so with 10,000 simulations we report $\mathbf{p < 0.0001}$.

5. Report the p-value. Make a decision on whether the results are statistically significant, and state your conclusion.

Since our p-value is less than 0.0001, which is well below the standard 0.05, we will reject the null hypothesis. We therefore conclude that the data provide enough evidence that the proportion of UFO sightings reported as spherical differs from 0.75.

Part 4: Extra Practice

Practice 4.1: Exploring the `ptail()` Function

In this exercise, we will practice using some basic R coding to calculate a p-value without using the `ptail()` function.

1. Consider the null distribution and point estimate from the AP Multiple Choice exercise. Using the `<=` operator, write a single line of code that will show which values in the null distribution are less than or equal to the point estimate. Run this code and observe what the values look like. Save this result to a vector. *Hint*: don't overthink this part. Just write the inequality as you would in math class.

```r
true_false_total <- null_dist_ap <= p_hat_ap
```

2. Using your vector from above, count how many values from the null distribution are less than or equal to the point estimate. Use the `sum()` function for this. *Hint*: `sum()` on a vector of `TRUE`/`FALSE` values counts the `TRUE`s, because R treats `TRUE` as 1 and `FALSE` as 0.

```
sum_true <- sum(true_false_total)
```

3. Using your result from part 2, find the proportion of values in the null distribution that are less than or equal to the point estimate. Compare to the p-value from `ptail()` in Practice 2.2.

```
prop_obs_lessthan_phat <- sum_true / length(null_dist_ap)
prop_obs_lessthan_phat
```

```
[1] 0.4721
```

Practice 4.2: UFO Shapes

UFOs, or Unidentified Flying Objects, have become commonplace in media and popular culture. Some witnesses claim that the objects have a cylindrical shape, while others believe they are more spherical. Load the data `ufo_data.csv` using the function `read.csv()`. The file is in the course materials folder; save it to your working directory (the same folder as this file) before running `read.csv()`. The data have already been filtered to include only sightings described as spheres or cylinders.

Run a hypothesis test to check whether the proportion of sightings described as spherical differs from 0.75.

Answers:

1. State the null and alternative hypotheses in words and in symbols.
   - Null: The proportion of UFO sightings described as spherical is 0.75.
   - Alternative: The proportion of UFO sightings described as spherical is different from 0.75.
   - $$H_0 : \pi = 0.75$$
   - $H_a : \pi \neq 0.75$
2. Calculate the observed statistic of interest.

```r
library(SDS1000)
ufo_data <- read.csv("ufo_data.csv")

p_hat_sphere <- get_proportion(ufo_data$shape, "sphere")
p_hat_sphere
```

```
   sphere
0.8076462
```

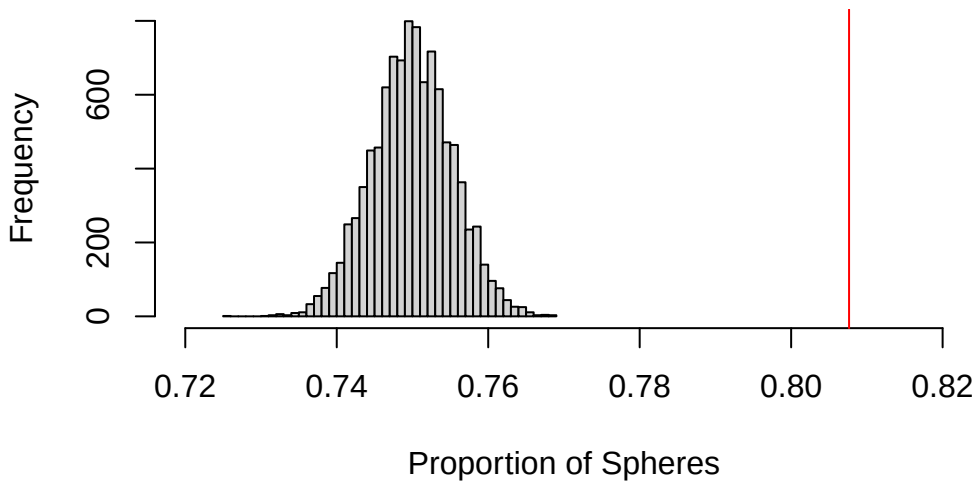
3. Create a null distribution in R using `do_it()` and `rflip()`. Plot the null distribution, and add a vertical red line at your point estimate.

```r
# Number of UFO sightings to simulate
n_ufos <- 6670

# Set the seed so everyone gets the same simulated results
set.seed(1000)

# Simulate the null distribution
null_dist_sphere <- do_it(10000) * {
  rflip(n_ufos, 0.75) / n_ufos
}

# Visualize the null distribution
hist(null_dist_sphere, breaks = 50,
     main = "Null Distribution of Proportion of Spherical UFOs",
     xlab = "Proportion of Spheres", xlim = c(0.72, 0.82))
abline(v = p_hat_sphere, col = "red")
```



4. Calculate the p-value (probability of obtaining a result as or more extreme than what we observed) using the `ptail()` function.

Because this is a two-sided test, "as or more extreme" means as far from the null value **in either direction**. Our observed proportion is about 0.058 above 0.75, so a result 0.058 *below* 0.75 would be just as extreme. We find that distance, mark a cutoff that far out on each side of the null value, and add the two tail probabilities.

```
null_val <- 0.75

# Distance between the observed proportion and the null value
margin <- abs(p_hat_sphere - null_val)

# Upper tail: as far above 0.75 as we observed, or farther
upper <- ptail(null_val + margin, null_dist_sphere, lower.tail = FALSE)

# Lower tail: the same distance below 0.75, or farther
lower <- ptail(null_val - margin, null_dist_sphere, lower.tail = TRUE)

pv <- lower + upper
pv
```

```
[1] 0
```

None of our 10,000 simulated proportions were as extreme as the one we observed, so `ptail()` returns 0. That does **not** mean the p-value is exactly 0 (that would mean our result is impossible under the null). A simulation-based p-value can't go below 1 / (number of simulations), so with 10,000 simulations we report $\mathbf{p < 0.0001}$.

5. Report the p-value. Make a decision on whether the results are statistically significant, and state your conclusion.

Since our p-value is less than 0.0001, which is well below the standard 0.05, we will reject the null hypothesis. We therefore conclude that the data provide enough evidence that the proportion of UFO sightings reported as spherical differs from 0.75.

Practice 4.2: Hypothesis Testing for Difference in Two Means /Caffeine Taps

A sample of college students was asked to tap their fingers at a rapid rate. The sample was then divided at random into two groups of ten students each. Each student drank the equivalent of about two cups of coffee, which included about 200 mg of caffeine for the students in one group but was decaffeinated coffee for the second group. After a two-hour period, each student was tested to measure finger tapping rate (taps per minute). The goal of the experiment was to determine whether caffeine produces an increase in the average tap rate.

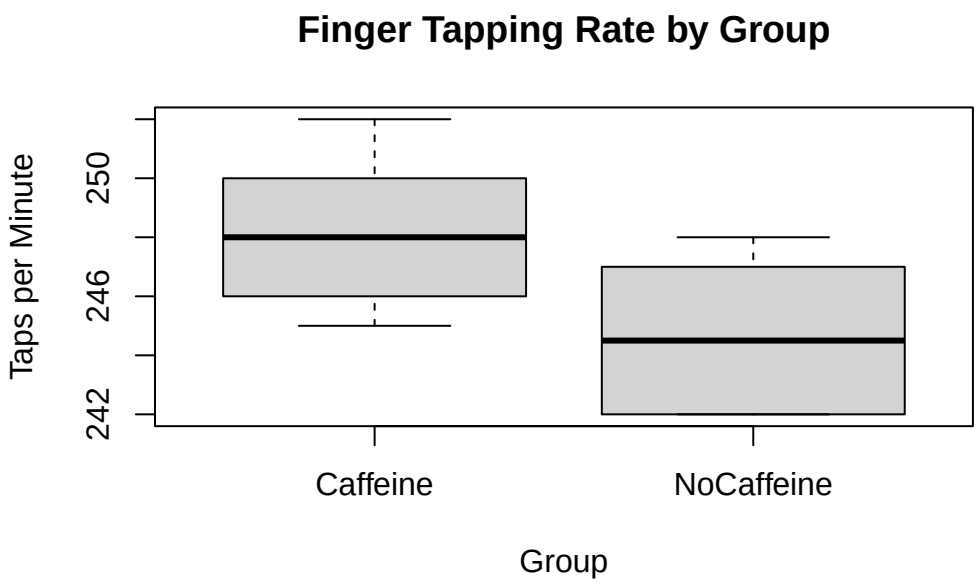
1. Write the null hypothesis and alternative hypothesis in words and in symbols.

    - Null: There is no difference in the average tapping rate between the caffeinated and decaffeinated groups.
    - Alternative: The average tapping rate is greater for the caffeinated group than for the decaffeinated group.
    - $H_0 : \mu_c - \mu_{nc} = 0$
    - $H_a : \mu_c - \mu_{nc} > 0$

2. Create a boxplot to describe tap rates for Caffeine versus No Caffeine.

```
library(Lock5Data)
library(SDS1000)
data(CaffeineTaps)

boxplot(Taps ~ Group, data = CaffeineTaps,
        main = "Finger Tapping Rate by Group",
        xlab = "Group", ylab = "Taps per Minute")
```



3. Find summary statistics for the number of taps in the `Caffeine` and `NoCaffeine` groups. You might find the function `mosaic::favstats()` useful. *Note*: you can search online for this function's arguments.

```
mosaic::favstats(Taps ~ Group, data = CaffeineTaps)
```

```
       Group min    Q1 median     Q3 max  mean       sd  n missing
1   Caffeine 245 246.5  248.0 250.00 252 248.3 2.213594 10       0
2 NoCaffeine 242 242.5  244.5 246.75 248 244.8 2.394438 10       0
```

4. Split the `Taps` variable into two vectors, one for the `Caffeine` group and one for the `NoCaffeine` group.

```
taps_caff   <- CaffeineTaps$Taps[CaffeineTaps$Group == "Caffeine"]
taps_nocaff <- CaffeineTaps$Taps[CaffeineTaps$Group == "NoCaffeine"]
```

5. Compute the observed statistic (the difference in mean taps between the two groups).

```
diff_caffeine <- mean(taps_caff) - mean(taps_nocaff)
diff_caffeine
```

```
[1] 3.5
```

6. Create the null distribution.

a. Combine the two groups into a single vector.

```
combined_sample <- c(taps_caff, taps_nocaff)
```

b. Create the null distribution using the `do_it()` function. Each time, shuffle the combined vector, split it back into two groups of the original sizes, and find the difference in means.

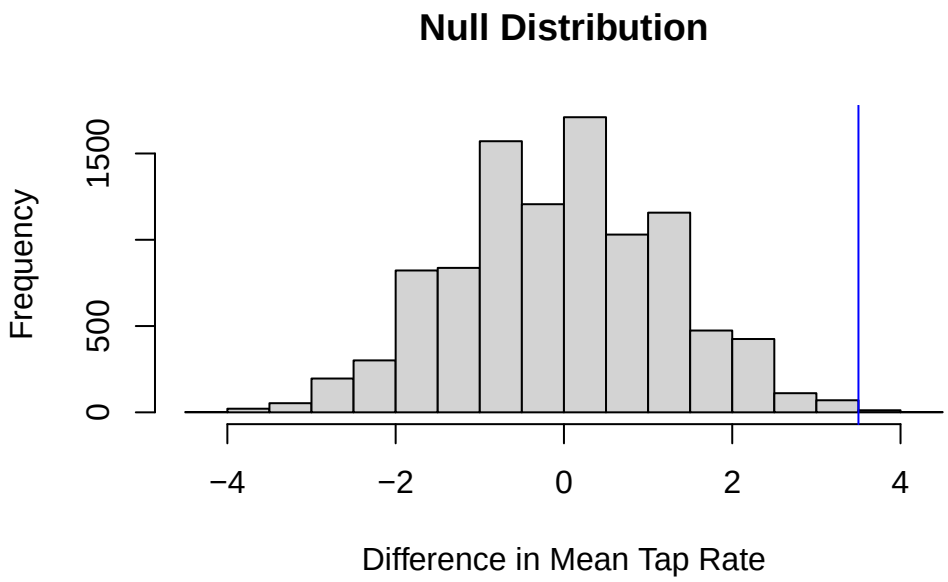
```
n_caff  <- length(taps_caff)
n_total <- length(combined_sample)

# Set the seed so everyone gets the same simulated results
set.seed(1000)

null_dist_caff <- do_it(10000) * {
  shuffled_sample <- sample(combined_sample)
  shuff_caff   <- shuffled_sample[1:n_caff]
  shuff_nocaff <- shuffled_sample[(n_caff + 1):n_total]
  mean(shuff_caff) - mean(shuff_nocaff)
}
```

c. Plot a histogram of the null distribution and show the line of the observed mean difference using the `abline()` function.

```
hist(null_dist_caff,
     xlab = "Difference in Mean Tap Rate", main = "Null Distribution")
abline(v = diff_caffeine, col = "blue")
```



7. Calculate the p-value.

```
p_value <- ptail(diff_caffeine, null_dist_caff, lower.tail = FALSE)
p_value
```

```
[1] 0.0026
```

8. Make a decision and state your conclusion.

Since our p-value is less than 0.05, we will reject the null hypothesis. We therefore have enough evidence that drinking caffeinated coffee produces an increase in the average tap rate compared to those who drink decaffeinated coffee.

Part 4: Extra Practice

Practice 3.1: Caffeine Taps

A sample of college students was asked to tap their fingers at a rapid rate. The sample was then divided at random into two groups of ten students each. Each student drank the equivalent of about two cups of coffee, which included about 200 mg of caffeine for the students in one

group but was decaffeinated coffee for the second group. After a two-hour period, each student was tested to measure finger tapping rate (taps per minute). The goal of the experiment was to determine whether caffeine produces an increase in the average tap rate.

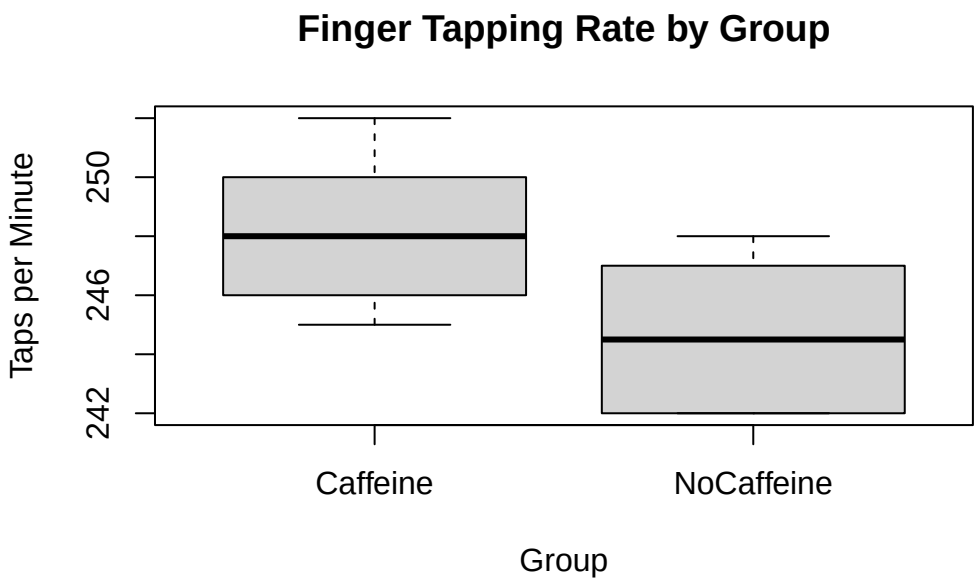
1. Write the null hypothesis and alternative hypothesis in words and in symbols.

    - Null: There is no difference in the average tapping rate between the caffeinated and decaffeinated groups.
    - Alternative: The average tapping rate is greater for the caffeinated group than for the decaffeinated group.
    - $H_0 : \mu_c - \mu_{nc} = 0$
    - $H_a : \mu_c - \mu_{nc} > 0$

2. Create a boxplot to describe tap rates for Caffeine versus No Caffeine.

```
library(Lock5Data)
library(SDS1000)
data(CaffeineTaps)

boxplot(Taps ~ Group, data = CaffeineTaps,
        main = "Finger Tapping Rate by Group",
        xlab = "Group", ylab = "Taps per Minute")
```



3. Find summary statistics for the number of taps in the `Caffeine` and `NoCaffeine` groups. You might find the function `mosaic::favstats()` useful. *Note*: you can search online for this function's arguments.

```
mosaic::favstats(Taps ~ Group, data = CaffeineTaps)
```

```
       Group min    Q1 median     Q3 max  mean       sd  n missing
1   Caffeine 245 246.5  248.0 250.00 252 248.3 2.213594 10       0
2 NoCaffeine 242 242.5  244.5 246.75 248 244.8 2.394438 10       0
```

4. Split the `Taps` variable into two vectors, one for the `Caffeine` group and one for the `NoCaffeine` group.

```
taps_caff   <- CaffeineTaps$Taps[CaffeineTaps$Group == "Caffeine"]
taps_nocaff <- CaffeineTaps$Taps[CaffeineTaps$Group == "NoCaffeine"]
```

5. Compute the observed statistic (the difference in mean taps between the two groups).

```
diff_caffeine <- mean(taps_caff) - mean(taps_nocaff)
diff_caffeine
```

```
[1] 3.5
```

6. Create the null distribution.

a. Combine the two groups into a single vector.

```
combined_sample <- c(taps_caff, taps_nocaff)
```

b. Create the null distribution using the `do_it()` function. Each time, shuffle the combined vector, split it back into two groups of the original sizes, and find the difference in means.

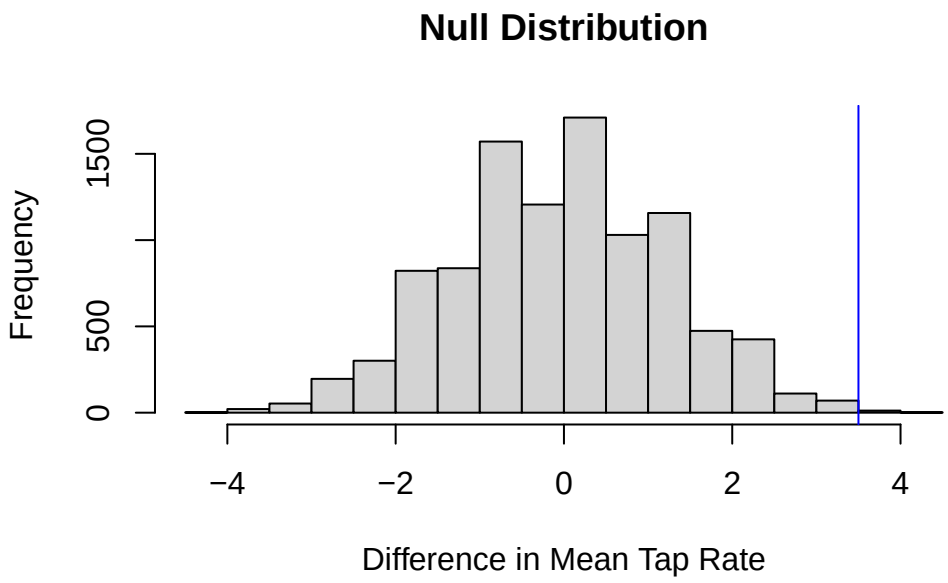
```
n_caff  <- length(taps_caff)
n_total <- length(combined_sample)

# Set the seed so everyone gets the same simulated results
set.seed(1000)

null_dist_caff <- do_it(10000) * {
  shuffled_sample <- sample(combined_sample)
  shuff_caff   <- shuffled_sample[1:n_caff]
  shuff_nocaff <- shuffled_sample[(n_caff + 1):n_total]
  mean(shuff_caff) - mean(shuff_nocaff)
}
```

c. Plot a histogram of the null distribution and show the line of the observed mean difference using the `abline()` function.

```
hist(null_dist_caff,
     xlab = "Difference in Mean Tap Rate", main = "Null Distribution")
abline(v = diff_caffeine, col = "blue")
```



7. Calculate the p-value.

```
p_value <- ptail(diff_caffeine, null_dist_caff, lower.tail = FALSE)
p_value
```

```
[1] 0.0026
```

8. Make a decision and state your conclusion.

Since our p-value is less than 0.05, we will reject the null hypothesis. We therefore have enough evidence that drinking caffeinated coffee produces an increase in the average tap rate compared to those who drink decaffeinated coffee.